

JYOT RAVAL

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EDUCATION

Pandit Deendayal Energy University (PDEU)

CGPA: 8.75 (Jun. 2025)

B.Tech in Information and Communication Technology

Expected Graduation : Jun. 2027

Relevant Coursework: Data Structures and Algorithms, Object Oriented Programming, Database Systems

EXPERIENCE

Machine Learning Intern

Nov 2025 – Mar 2026

Central Tickets

Hybrid

- Developed a **RAG-based chatbot** with automated data scraping after each website update, ensuring up-to-date knowledge and providing **source citations** for user verification.
- Implemented **Model Context Protocol (MCP)** to enable autonomous booking, cancellation, and refund workflows, allowing chatbot to safely interact with internal tools and APIs, **resolved 50+ user queries weekly**, reducing repetitive customer calls and email tickets.
- Designed an **NLP-based clustering pipeline** Designed an NLP clustering pipeline in Python to group similar customer email issues, which reduced manual triage time by 30% and improved routing accuracy for support teams.
- Processed **200,000+ historical customer queries** to extract the **top 20 recurring problem categories**, enabling structured FAQs and faster issue resolution.

PROJECTS

RepoViz: Python Codebase Hierarchical Visualizer

Sept 2025 – Nov 2025

Tech Stack: Python, NetworkX, Graphviz

[Source Code](#)

- Designed a static analysis pipeline scanning **50+ Python files per minute** to extract module dependencies and build detailed graphs using NetworkX.
- Achieved **100% parsing accuracy** through extensive test coverage, including support for custom and non-standard import patterns.
- Delivered both a **CLI and PyQt5-based GUI** built on a shared core pipeline, improving usability for both terminal and desktop users.

Contest Unify: Unified CP Contest Tracker & AI Coach

May 2025 – July 2025

Tech Stack: Django, PostgreSQL, GitHub Actions, Gemini

[Source Code](#)

- Built a backend to unify contest data from **5+ CP platforms** using RESTful APIs, GraphQL, and HTML scraping with BeautifulSoup, stored data with deduplication in **PostgreSQL (Supabase)**.
- Automated scheduled fetching of **20+ contests per run** using GitHub Actions with logging and failure alerts, ensuring consistent and reliable updates.
- Developed an AI assistant that analyzed user handles to generate **personalized contest recommendations and performance summaries**, tested by **15+ early users**.

TECHNICAL SKILLS

Programming and Scripting Languages: Python, Java, C/C++

Web & Frameworks: Django, FastAPI, Flask, HTML, CSS, JavaScript

Databases: PostgreSQL, SQLite, Pinecone, FAISS

Tools: Git, GitHub, GitHub Actions, VS Code, Linux CLI

Core CS: DSA, OOPs, System Design

COMPETITIVE PROGRAMMING PROFILES

Knight @ Leetcode: [jyotraval](#) (Max. 1973)

3 Star @ CodeChef: [jyotraval](#) (Max. 1655)

Pupil @ Codeforces: [jyotraval](#) (Max. 1300)